

GRADE RELATED PAPER

Implementation of the GRADE adolopment framework was perceived as appropriate, acceptable, and feasible in a national guideline program in the Philippines: a mixed-methods study

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Abstract

Background: The Grading of Recommendations Assessment Development and Evaluation (GRADE) Adolopment framework provides a structured approach for adapting and developing clinical practice guidelines; however, evidence on its implementation in low- and middle-income countries (LMICs) is limited. This study evaluated the implementation of the GRADE Adolopment framework in the development of the Philippine Periodic Health Examination (PHEX) screening guidelines and identified perceived barriers, enablers, and areas for improvement.

Study Design and Setting: We conducted a sequential exploratory mixed-methods study involving a document review, key informant interviews ($n = 14$), and a survey ($n = 109$) among members of the PHEX guideline task force in the Philippines. Thematic analysis and integration of quantitative and qualitative findings were used to assess implementation outcomes: appropriateness, acceptability, and feasibility.

Results: Implementation required contextual modifications to the GRADE Adolopment process, including de novo development of evidence-to-decision tables and use of rapid reviews. Most survey respondents found the framework appropriate (93.6%), acceptable (99.1%), and feasible (96.3%). Qualitative findings indicated that structured methods and alignment with local policy needs contributed to perceived appropriateness, while transparency supported acceptability. Enablers included technical guidance, prior experience, and high-quality evidence access. Key barriers included limited familiarity with the framework, lack of standardized processes, and insufficient resources and capacity.

Conclusion: The GRADE Adolopment framework can be effectively implemented in LMIC settings with appropriate adaptations. Investments in capacity-building, interest holder engagement, and resource allocation are essential to sustain its use and improve the development of evidence-based clinical practice guidelines in the Philippines and similar contexts. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: GRADE Adolopment; Clinical practice guidelines; Mixed-methods research; Implementation outcomes; Low- and middle-income countries

1. Introduction

Clinical practice guidelines (CPGs) serve as vital tools for promoting evidence-based decision-making in health systems. By translating evidence into actionable recommendations, CPGs have the potential to reduce practice variation,

improve health outcomes, and inform policy and financing decisions [1,2]. In low- and middle-income countries (LMICs) such as the Philippines, the development of trustworthy, contextually appropriate guidelines is essential to improve quality of care, and optimize the use of limited health resources.

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Plain Language Summary

Clinical practice guidelines help doctors and health systems make decisions about screening, treatment, and disease prevention. Developing these guidelines from scratch requires substantial time, expertise, and resources. One approach to make guideline development more efficient is the GRADE Adolopment framework. This study evaluated how the GRADE Adolopment framework was implemented in the Philippines during the development of national guidelines for preventive health screening through the PHEX project. Using interviews, document review, and a survey of guideline developers, the study examined whether the framework was appropriate, acceptable, and feasible in this setting. Most participants reported that the framework improved transparency, structure, and collaboration in guideline development. However, implementation required significant technical expertise, training, and coordination. Limited familiarity with the framework and gaps in local evidence were identified as important challenges. These findings suggest that the GRADE Adolopment framework can be used successfully in LMICs, but sustained implementation requires investments in training, methodological support, and stronger interest holder engagement.

Recognizing this need, the Philippine Department of Health (DOH) institutionalized a National Practice Guideline Program to produce high-quality, evidence-based guidelines for various health priorities, including preventive screening [3]. In support of this initiative, Philippine Department of Health commissioned a team to update the Philippine Periodic Health Examination (PHEX) guidelines, the national guidance on screening interventions for apparently healthy individuals.

Although guideline adaptation methods have been developed to streamline the production of CPGs, traditional frameworks often remain resource-intensive and require technical expertise that is not always available in LMIC settings. The GRADE Adolopment framework, introduced by the GRADE Working Group, offers a more flexible approach by integrating evidence appraisal, contextualization, and decision-making through the GRADE evidence-to-decision (EtD) framework [4,5]. Despite its growing adoption, few studies have described or evaluated how GRADE Adolopment is implemented in real-world LMIC contexts [6].

In 2021, the GRADE Adolopment framework was used to update the PHEX guidelines in the Philippines. This presented a unique opportunity to examine the framework's implementation in a national program involving multiple task forces and a diverse group of interest holders. This article describes the Philippine experience in implementing the GRADE Adolopment framework for CPG development, evaluates its perceived appropriateness, acceptability, and feasibility from different interest holders' perspectives, and offers lessons for future guideline developers and health authorities in similar settings.

2. Methods

2.1. Study design

This study used an exploratory sequential mixed-methods design (QUAL→QUAN), in which qualitative

findings were used to inform and develop quantitative survey questions. A mixed-methods approach was chosen to capture both in-depth contextual experiences of implementation and the distribution of perceptions across a larger group of guideline developers. The qualitative component involved document review and semistructured interviews, while the quantitative component utilized a cross-sectional survey. Both data types were collected and analyzed to assess perceptions of appropriateness, acceptability, and feasibility.

2.2. Study setting

The study was conducted in the context of the development of the PHEX guidelines, a national initiative led by the Department of Health of the Philippines. Originally developed in 2005, the PHEX guidelines cover a broad range of screening conditions for both pediatric and adult populations. The 2021–2023 effort represents the first major update to the guidelines, prompted by the need to reflect current evidence and health priorities. Due to the enormity of the topics covered, the PHEX guidelines were developed in three phases. Phase 1 (2021–2022) aimed to provide evidence-based recommendations for screening 16 prioritized health conditions through a single task force, while phases 2 and 3 were an expansion to include 12 topic specific-task forces [7].

Following prescribed national CPG protocol, each task force involved members assigned to specific roles: the Steering Committee provided strategic oversight and policy direction; the Technical Working Group (TWG) conducted evidence appraisal and drafted the EtD tables and recommendations; and the Consensus Panel reviewed, discussed, and finalized the recommendations based on the presented evidence [7,8]. In total, over 200 participations were recorded across phases 1 and 2; however, this includes individuals who may have served multiple roles or participated in more than one task force. The detailed process of development and the developed PHEX recommendations have been previously published [7].

What is new?**Key findings**

- GRADE Adolopment was perceived as appropriate, acceptable, and feasible when implemented in a national clinical practice guideline program in a low- and middle-income country (LMIC).

What this adds to what is known?

- This study provides empirical mixed-methods evidence on the real-world implementation of GRADE Adolopment in an LMIC setting, addressing a key gap in the guideline methodology literature.

What is the implication and what should change now?

- Health systems especially in LMIC may consider the GRADE Adolopment framework to support development of CPGs.
- Strengthening methodological capacity and technical support may facilitate its implementation in guideline programs.

2.3. Study population

The study involved two groups of participants: (1) key informants purposively selected from different roles and experiences involved in the guideline development process and (2) members of the guideline development task forces who were invited to participate in an online survey.

2.4. Data collection**2.4.1. Qualitative**

The qualitative data collection involved a document review of relevant project materials and semistructured interviews with key informants to explore their experiences, perceptions, and contextual insights regarding the implementation of the GRADE Adolopment process. Materials reviewed included documents generated during the guideline development process, such as evidence summaries, EtD tables, and technical reports. Interviews were audio-recorded, transcribed verbatim, and analyzed.

2.4.2. Quantitative

A structured survey was administered to guideline task force members to assess perceptions of the appropriateness, acceptability, and feasibility of the GRADE Adolopment framework. The survey instrument was informed by findings from the qualitative phase and guided by Proctor et al's implementation outcomes framework constructs [9]. Themes

identified from the document review and key informant interviews—particularly those related to appropriateness, acceptability, feasibility, barriers, enablers, and areas for improvement—were translated into survey domains and questionnaire items. Prior to administration, the questionnaire was pretested to improve clarity and relevance. Perceptions of appropriateness, acceptability, and feasibility were measured using Likert-scale items ranging from strongly disagree to strongly agree, while barriers and enablers were assessed using checklist-style items in which respondents selected factors they perceived as influencing implementation.

2.5. Analysis**2.5.1. Qualitative**

Qualitative data were analyzed using content analysis guided by Proctor et al's implementation outcomes framework and managed using MAXQDA [9,10]. Manifest content analysis was used to describe visible patterns and participant expressions. Both deductive and inductive coding approaches were used. Deductive coding was guided by predefined implementation outcomes (appropriateness, acceptability, and feasibility), while inductive coding allowed identification of emergent themes such as enablers, barriers, and areas for improvement. Initial coding was conducted by the first author (I.T.C.) and a trained research analyst. Discrepancies in coding or interpretation were resolved by consensus, and themes were analyzed post-hoc by participant role. Thematic saturation was considered achieved when additional interviews yielded no new themes related to the implementation of the GRADE Adolopment framework. The study focused on three core implementation outcomes [9,11]—appropriateness, acceptability, and feasibility—and explored perceived enablers, barriers, and areas for improvement from the perspective of task force members involved in the guideline development process.

2.5.2. Quantitative

Descriptive statistics, including frequencies, percentages, means, and standard deviations, were computed to summarize respondent characteristics and responses to items on perceived appropriateness, acceptability, and feasibility. Responses were further analyzed according to participant role (Steering Committee, TWG, Consensus Panel) to explore potential differences in perceptions. Stata 15.1 was used to analyze the data [12].

2.6. Mixed methods integration

Qualitative and quantitative findings were integrated at two stages of the mixed-methods design. First, integration occurred at the design stage, where themes identified during the qualitative phase informed the development of the survey instrument administered in the quantitative phase. Second, integration occurred at the interpretation stage,

where qualitative themes and quantitative survey findings were compared and interpreted together to examine convergence, divergence, and complementarity.

The analysis was guided by Proctor et al's implementation outcomes framework, focusing on the constructs of appropriateness, acceptability, and feasibility. The research team conducted iterative discussions to synthesize findings from both data sources and develop an integrated interpretation of the implementation experience.

The University of the Philippines Manila Research Ethics Board (2021-558-01) exempted the study protocol from ethical review because it involved minimal risk and focused on evaluating program implementation. All participants provided digital informed consent.

3. Results

A total of 14 PHEX phase 1 task force participated in the interviews. Of 175 target respondents, 109 (62.3%) participated in the survey. 18 documents were reviewed.

3.1. Document review

A total of 16 evidence summaries with EtD tables and two internal reports were reviewed. EtD tables were developed de novo for all topics rather than adapted from source guidelines. However, there was inconsistency in the quality and completeness of documentation across task forces. Gaps were identified in role definition, declaration of conflicts of interest, and documentation of decision points.

3.2. Key informant interview and survey respondent characteristics

The key informants had a mean age of 41 years, with most being medical doctors (57.1%) or allied health professionals (35.7%). Nine of 14 were part of the technical working group, and half (7 of 14) had no prior experience with guideline development (Table 1). Only 28.5% had prior knowledge of GRADE. The survey respondents were also predominantly medical doctors (92.7%) with a mean age of 43.9 years (range: 25–72). They represented various roles in the guideline development process, including technical working group members (52.3%), consensus panel members (28.4%), and steering committee members (19.3%). Most had prior experience in clinical practice guideline development (72.0%), reported familiarity with the GRADE approach (67.9%), and had training or exposure to evidence-based health care (71.6%) (Table 1).

According to the interviews, participants consistently found the framework appropriate, citing its comprehensiveness and value in incorporating interest holder perspectives (Table 2). It was viewed as acceptable due to its structured and transparent nature. However, the feasibility was

challenged by the need for highly skilled personnel, time, and administrative coordination. Key enablers included access to high-quality source guidelines, strong guidance, and existing methodological expertise. Barriers included unfamiliarity with the framework, lack of standardization of the methods, limited funding, and insufficient engagement from patient representatives and consensus panel members. Areas for improvement included expanded training especially on how to interpret the evidence using GRADE, interest holder engagement, and streamlined processes.

Table 3 presents a joint display integrating qualitative themes and quantitative survey findings. Survey results complemented the qualitative findings by quantifying perceptions of appropriateness, acceptability, and feasibility, with most respondents reporting the framework as appropriate (93.6%), acceptable (99.1%), and feasible (96.3%). See Supplementary.

Survey findings further clarified these perceptions. A majority agreed that the framework was comprehensive (88.1%), allowed localization (79.8%), and supported multisectoral collaboration (87.1%). Most agreed that it improved guideline quality (91.7%) and promoted transparency (91.8%). Despite the overall feasibility, a third of respondents found it overwhelming or required substantial learning time. See supplementary for detailed survey results.

Top enablers included technical guidance and access to quality evidence. The most cited barriers were unfamiliarity (44.0%) and evidence gaps (43.1%). Capacity-building was the most frequently recommended area for improvement (63.3%). See supplementary.

4. Discussion

This study found that the GRADE Adolopment framework was widely regarded by PHEX guideline developers as appropriate, acceptable, and feasible. Participants valued its structured and transparent processes, the explicit use of the EtD framework, and its ability to capture interest holder values and improve the quality of recommendations. These findings align with previous reports that emphasize the GRADE framework's clarity and its ability to facilitate transparent decision-making [6,13,14].

Although the overall structure of the GRADE Adolopment framework was maintained, implementation required several contextual adaptations. In particular, EtD tables were developed de novo for all topics rather than adapted from existing source guidelines, reflecting the need to tailor the framework to the available evidence and local context. The documents also reflected active interest holder participation, use of standardized templates, and structured consensus discussions.

While the framework was considered feasible, its implementation was resource—and expertise-intensive. Prior

Table 1. Characteristics of key informant interviewees and survey respondents

Characteristic	Key informants (n = 14) (%)	Survey respondents (n = 109) (%)
Mean age [SD] (years)	41 [8.0]	43.9 [11.6]
Profession		
Medical doctor	8 (57.1)	101 (92.7)
Allied health professionals ^b	5 (35.7)	6 (5.5)
Others	1 (7.1)	2 (1.8)
CPG role		
Steering committee	2 (14.3)	21 (19.3)
Technical Working Group (TWG)	9 (64.3)	57 (52.3)
Consensus panel	3 (21.4)	31 (28.4)
Prior experience in CPG development		
None	7 (50.0)	44 (40.4)
1–3 guidelines	6 (42.9)	53 (48.6)
≥4 guidelines	1 (7.1)	12 (11.0)
Prior knowledge of GRADE	4 (28.5)	74 (67.9)
Self-reported training background ^a		
Clinical epidemiology	6 (42.9)	29 (26.6)
Evidence-based medicine	4 (28.6)	78 (71.6)
Public health/health policy	2 (14.3)	23 (21.1)
Health research methods	3 (21.4)	39 (35.8)
Health technology assessment	1 (7.1)	5 (4.6)

CPG, clinical practice guideline; GRADE, Grading of Recommendations Assessment Development and Evaluation.

^a Multiple responses allowed.

^b Nurse, speech pathologist, physical therapist, pharmacist.

studies have noted the high technical demands of guideline adaptation methods, even those designed for efficiency [13,15]. When compared with experiences in other countries, our findings were consistent. Earlier adaptation frameworks such as ADAPTE were also reported as time- and resource-intensive, taking up to 18 months to 3 years. The guideline on Rheumatoid Arthritis for the Eastern Mediterranean Region, with close support from the GRADE Working Group, achieved much faster production—80 EtDs in 4 months [16]—reflecting stronger institutional capacity and prior methodological expertise. Similar to our context, other LMICs such as Tunisia [6] and Pakistan [15] reported barriers including lack of trained personnel, methodological unfamiliarity, limited patient involvement, and challenges in sustaining funding.

Successful application in the PHEX project required strong leadership, extensive coordination, and technically skilled staff, especially in evidence appraisal and recommendation formulation. The most frequently cited area for improvement was capacity building especially on evidence synthesis for the technical working group and interpreting GRADE and EtD for panel member. Respondents emphasized the need for onboarding, training, and continuous mentorship. Broader participation from patients and implementers was also lacking but deemed essential for future CPG adaptation endeavors. Differences in familiarity with GRADE and roles across participants may have influenced both qualitative insights and survey responses, affecting the integration of findings.

Table 2. Summary of key themes from key informant interviews

Constructs	Themes	Illustrative quotes
Implementation outcomes		
Appropriateness	Structured and comprehensive; emphasizes values and preferences	“I really appreciated the evidence-to-decision framework (because) it really explicitly points out what are the items that you need to really consider...”—steering committee member “It was actually based on their own value system... you ask them to grade... so <i>maganda siyang metric (it was a good metric)</i> ...”—Consensus panel member
Acceptability	Promotes learning; encourages reflection and trust; transparent process	“[The process] is an eye-opener <i>para sa akin</i> (for me) and I learned a lot and very encouraging for me to know that this is going on”—CP member
Feasibility	Technically demanding; requires skilled facilitation; improved efficiency	“Since it’s a new methodology... only a few people are skilled enough to practice it.”—TWG member “It allows you to complete a guideline in just a fraction of the time...”—TWG member
Enablers and barriers		
Enablers	Technical guidance; core team support; access to good evidence	“The work was made easier by the assistance provided primarily by the project (core) team.”—TWG member
Barriers	Lack of familiarity; technical complexity; insufficient local evidence	“Yeah, the experience it was challenging because I am unaware of the processes, the specific processes involved.”—TWG member

Themes derived from qualitative content analysis of 14 key informant interviews. Quotes are anonymized. Translations provided for Filipino/Taglish statements.

TWG, Technical Working Group.

Table 3. Integration of qualitative interview findings and quantitative survey results on the implementation of the GRADE Adolopment framework

Implementation outcome/theme	Qualitative findings (interviews) <i>n</i> = 14	Quantitative findings (survey) <i>n</i> = 109
Appropriateness	Participants described the framework as structured and systematic for evaluating evidence and supporting recommendation development.	Majority of respondents (93.6%) perceived the framework as appropriate for the guideline development process.
Acceptability	Interviewees noted improved transparency and structured discussion among task force members.	Most respondents (99.1%) reported the framework was acceptable to guideline developers.
Feasibility	Participants noted that implementation required technical expertise and familiarity with the framework.	Most respondents (96.3%) reported the framework was feasible, though some identified training needs.
Enablers	Technical guidance, prior experience with evidence-based methods, and collaboration within the guideline task force.	Respondents similarly identified training, methodological support, and access to evidence as enablers.
Barriers	Limited familiarity with the framework, lack of standardized documentation, and resource constraints.	Respondents also identified limited experience and resource limitations as barriers.

GRADE, Grading of Recommendations Assessment Development and Evaluation.

5. Conclusion

This study provides empirical evidence on the implementation of the GRADE Adolopment framework in a national guideline program in the Philippines. The mixed-methods design, integrating document review, key informant interviews, and survey data, enabled triangulation of findings and a comprehensive assessment of implementation experiences. The framework was perceived as appropriate, acceptable, and feasible in this setting. However, successful implementation required substantial technical capacity, resources, and contextual adaptation. A key limitation is that the findings reflect the Philippine context and may not be fully generalizable to other settings. Strengthening methodological capacity and stakeholder engagement may support broader implementation of the framework in similar contexts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT by OpenAI in order to support the organization of content, language refinement, and formatting during manuscript preparation. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRedit authorship contribution statement

Ian Theodore Cabaluna: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Katrice Biños:** Investigation, Data curation. **Rosally Zamora:** Investigation, Data curation. **Angela Castrillo:** Project

administration, Investigation, Data curation. **Cynthia Cordeiro:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Marissa Alejandria:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Lia Palileo-Villanueva:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Leonila Dans:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

I.T.C. reports a relationship with Republic of the Philippines Department of Health that includes: funding grants. The other authors declare no competing interests.

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112319>.

Data availability

Data will be made available on request.

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